

Sodium Bicarbonate for Critically Ill Adults with Metabolic Acidosis and Shock

The SODa-BIC Randomized Clinical Trial

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Study Design: Multinational, Pragmatic, Adaptive, Double-Blind, Placebo-Controlled Randomized Clinical Trial (55 ICUs, 7 Countries, N = 500)

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Article Metadata & Trial Identification

Core Publication Details

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- **Coordinating Groups:** ANZICS Clinical Trials Group & ANZIC Research Centre, Monash University, Melbourne
- **Corresponding Author:** Prof. Ary Serpa Neto (Monash University / Austin Hospital)
- **Primary Funding:** National Health and Medical Research Council (NHMRC) of Australia

Scope & Research Framework

- **Clinical Setting:** 55 Intensive Care Units across 7 countries (Australia, New Zealand, Brazil, Canada, Japan, Ireland, Saudi Arabia)
- **Patient Population:** Adults in shock receiving continuous vasopressor infusion with metabolic acidosis ($\text{pH} < 7.30$, $\text{BE} \leq -4$ mmol/L)
- **Core Dilemma:** Prior trials (BICAR-ICU 2018, BICARICU-2 2025) suggested potential renal benefit but were open-label and limited to severe acidemia ($\text{pH} \leq 7.20$)
- **SODa-BIC Innovation:** First large-scale, double-blind, placebo-controlled pragmatic trial testing protocolized buffer therapy in shock

Executive Summary & Practice Bottom Line

CLINICAL QUESTION	PRIMARY RESULT	KEY BENEFIT / HARM	PRACTICE BOTTOM LINE
Does early, protocolized IV sodium bicarbonate infusion improve 30-day major adverse kidney events in critically ill adults with metabolic acidosis receiving vasopressors?	MAKE30 occurred in 40.2% (98/244) in Bicarbonate vs 39.4% (100/254) in Placebo (Adj Diff +1.2% pts; 95% CI: -7.1 to 9.4; P=0.78). No difference in mortality or RRT.	Physiological correction of pH achieved faster (recurrence 32% vs 56%), but NO benefit on vasopressor duration, AKI progression, or survival. Hypokalemia was slightly higher (1.6% vs 0%).	DO NOT CHANGE PRACTICE. Routine sodium bicarbonate is NOT recommended for moderate metabolic acidosis (pH < 7.30) in vasopressor-dependent shock.

Pathophysiological Rationale: Acidemia & Shock

The 'Acidemia-Shock' Vicious Cycle

1. Reduced Adrenoreceptor Responsiveness:

- Acidemia (pH < 7.20-7.30) impairs beta- and alpha-1 adrenergic signaling, blunting inotropy and vasoconstrictor response.

2. Myocardial Depression & Vasodilation:

- Intracellular acidosis decreases troponin calcium sensitivity, impairing stroke volume and arterial tone.

3. Renal Hypoperfusion & AKI:

- Renal vasoconstriction and systemic hypotension trigger tubular ischemia, oliguria, and early need for RRT.

The Bicarbonate Dilemma in Critical Care

• Theoretical Hazards of Sodium Bicarbonate:

- Paradoxical intracellular acidosis (CO₂ freely diffuses across cell membranes while HCO₃⁻ remains extracellular).
- Left shift of oxyhemoglobin dissociation curve (impaired tissue O₂ unloading).
- Hyperosmolarity, hyponatremia, volume overload, and hypocalcemia/hypokalemia.

• The Unresolved Evidence Gap:

- Surviving Sepsis Campaign 2021 recommends against NaHCO₃ for hypoperfusion-induced lactic acidemia with pH ≥ 7.20 (low quality evidence).
- Open-label trials had high risk of confirmation bias for RRT initiation.

Clinical Question & Structured PICO Framework

P	POPULATION	Critically ill adults (≥ 18 yr) admitted to ICU receiving continuous vasopressor infusion to correct hypotension, with metabolic acidosis: pH < 7.30 , Base Excess ≤ -4 mmol/L, PaCO ₂ ≤ 45 mmHg (unintubated) or ≤ 50 mmHg (intubated). Met within 2h of randomization.
I	INTERVENTION	Protocolized IV Sodium Bicarbonate (600 mmol/L in 5% Dextrose) starting at 100 mL/hr, titrated via mandatory ABG at 1h & 3h to target pH ≥ 7.30 and BE ≥ 0 mmol/L, stopped at 5 hours (maximum infusion window).
C	COMPARATOR	Identical Placebo (5% Dextrose in water without bicarbonate), matched in appearance, volume, starting rate (100 mL/hr), and blinded algorithmic titration for up to 5 hours.
O	OUTCOMES	Primary: Major Adverse Kidney Event within 30 days (MAKE30: all-cause death, receipt of RRT, or persistent renal dysfunction [creatinine $\geq 200\%$ baseline]). Secondary: 30d/90d mortality, vasopressor-free days, ICU-free days.

Trial Design & Methodological Architecture

Pragmatic & Adaptive

- **Adaptive Design:**
 - Pre-planned blinded sample size re-estimation after first 100 patients to ensure 90% power.
 - Interim safety review by independent DSMC after 125 patients.
- **Pragmatic Execution:**
 - Embedded in 55 routine ICUs across 7 countries with real-world clinician decision-making post-5 hours.

Robust Double-Blinding

- **Central Web Randomization:**
 - 1:1 ratio with permuted blocks of varying sizes.
 - Stratified by site, baseline pH (<7.25 vs ≥ 7.25), and creatinine (<1.7 vs ≥ 1.7 mg/dL).
- **Masking Integrity:**
 - Unblinded site preparer mixed D5W or NaHCO_3 ; bedside clinicians, patients, and outcome adjudicators completely blinded.

Target Population Focus

- **Vasopressor Enrichment:**
 - Targeted patients in shock where buffer therapy is most biologically plausible and clinically contested.
- **Broad Acidemia Spectrum:**
 - Unlike prior trials ($\text{pH} \leq 7.20$), included $\text{pH} < 7.30$, capturing real-world ICU practice where mild-moderate acidosis triggers buffer use.

Study Population & Eligibility Criteria

INCLUSION CRITERIA (All Met within 2h)

1. Age ≥ 18 years
2. Admitted to Intensive Care Unit (ICU)
3. Continuous Vasopressor Infusion to treat hypotension (Norepinephrine, Epinephrine, Vasopressin, etc.)
4. Documented Metabolic Acidosis on ABG:
 - Arterial pH < 7.30
 - Base Excess ≤ -4.0 mmol/L
 - PaCO₂ ≤ 45 mmHg (without invasive mechanical ventilation) OR ≤ 50 mmHg (with invasive mechanical ventilation)
5. Randomization initiated within 48h of ICU admission

KEY EXCLUSION CRITERIA

- **Renal Exclusion:** Current or planned RRT; baseline eGFR < 30 mL/min/1.73m² (CKD stage 4/5)
- **Specific Acidosis Types:** Diabetic Ketoacidosis (DKA); major digestive/urinary bicarbonate losses; known poisoning (methanol, ethylene glycol, aspirin)
- **Neurologic / Safety:** High risk of cerebral edema / acute brain injury; severe dysnatremia (Na < 120 or > 155); hypocalcemia; severe hypokalemia (K < 3.0)
- **Respiratory:** Severe pulmonary edema with PaO₂/FiO₂ < 100
- **Clinical:** Imminent/inevitable death; life expectancy < 30 days; ongoing NaHCO₃ therapy prior to enrollment

Intervention Protocol & 5-Hour Infusion Algorithm

Standardized 5-Hour Closed Titration Algorithm

START (0 - 1h)

100 mL / hr

Infusion started within 1h of randomization at 100 mL/hr in all patients.

EVALUATION 1 (Hour 1)

ABG Assessment

If pH < 7.30 or BE < 0: Maintain 100 mL/hr
If pH 7.30-7.35 & BE ≥ 0: Reduce to 25 mL/hr
If pH > 7.35 & BE > 0: Stop infusion

EVALUATION 2 (Hour 3)

ABG Assessment

If pH < 7.30 or BE < 0: Maintain 100 mL/hr
If pH 7.30-7.35 & BE ≥ 0: Reduce to 25 mL/hr
If pH > 7.35 & BE > 0: Stop infusion

COMPLETION (Hour 5)

Mandatory Stop

Infusion stopped permanently at 5 hours. Subsequent open-label NaHCO3 left to clinician discretion.

Trial Endpoints & Prespecified Hierarchy

PRIMARY ENDPOINT: MAKE30

Major Adverse Kidney Events by Day 30 (MAKE30):

A composite patient-centered endpoint defined as the occurrence of ANY of the following:

1. **Death from any cause** before hospital discharge or by Day 30.
2. **Receipt of Renal-Replacement Therapy (RRT)** within 30 days.
3. **Persistent Renal Dysfunction** (final inpatient creatinine closest to Day 30 $\geq$ 200% of baseline value).
 - Censored at hospital discharge or Day 30, whichever occurred first.
 - NIDDK/ADQI standard endpoint resilient to competing risks of death vs RRT.

SECONDARY & PROCESS ENDPOINTS

• Secondary Clinical Endpoints:

- Individual components of MAKE30 (Death, RRT, Persistent Dysfunction)
- RRT dependence at Day 30 (RRT within $\pm$ 10 days of Day 30)
- In-ICU death by Day 30 & In-hospital death by Day 90

• Physiological & Process Measures:

- Recurrence of metabolic acidosis within 7 days
- Incidence & stage of AKI (KDIGO criteria) within 7 days
- Vasopressor-free days, RRT-free days, & ICU-free days at Day 30
- Hospital-free days at Day 90
- Prespecified adverse events (hypokalemia, alkalemia, hyponatremia)

Statistical Analysis Plan & Power Estimation

Sample Size & Power Calculation

- **Baseline Assumption:** Control MAKE30 rate assumed to be 40% based on observational ICU data.
- **Effect Size Target:** 14 percentage point absolute reduction (from 40% to 26%), assuming ~70% of the effect seen in BICAR-ICU.
- **Power:** 90% power at two-sided $\alpha = 0.05$ required 470 patients (increased to 500 for dropouts).
- **Blinded Re-estimation:** Pre-planned sample size re-estimation after 100 patients confirmed the 500 target was adequate (cap set at 700).

Analytical Models & Strategy

- **Primary Analysis Population:** Modified Intention-to-Treat (mITT) including all randomized patients who did not withdraw consent.
- **Primary Model:** Mixed-effects generalized linear model with binomial distribution and identity link (site as random effect; stratification factors pH & creatinine as fixed effects).
- **Secondary Endpoints:** Continuous outcomes via mixed-effects median regression; survival to 30d/90d via shared-frailty Cox models.
- **Multiplicity Note:** Secondary endpoints reported with unadjusted 95% CIs (exploratory, no alpha control).

Figure 1: CONSORT Participant Flow Diagram

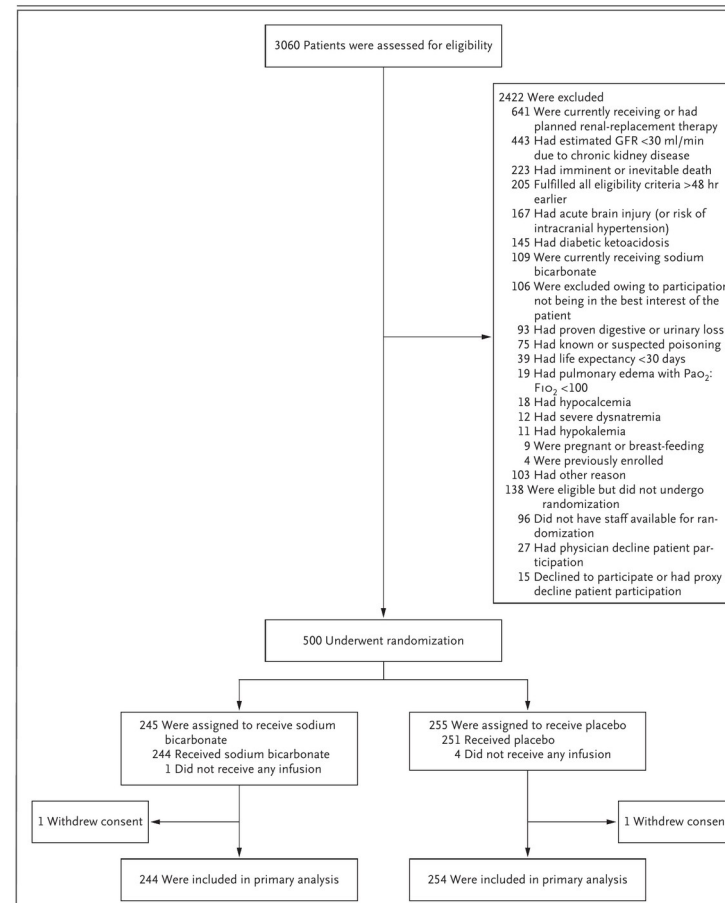


Figure 1. Enrollment, Randomization, and Follow-up of the Patients.

In settings that required prospective consent, no enrolled participants died before informed consent was obtained; thus, no exclusions related to informed-consent procedures occurred. Consent was withdrawn by 2 patients, which resulted in a modified intention-to-treat population of 498 patients. GFR denotes glomerular filtration rate, Fio₂ fraction of inspired oxygen, and Pao₂ partial pressure of arterial oxygen.

Participant Flow & Screening Appraisal

Critical Audit of Screening and Follow-up Integrity

High Screening Yield

3,060 patients assessed across 55 ICUs; 2,422 excluded (most common: baseline CKD eGFR <30 [443], ongoing/planned RRT [641], imminent death [223]).

Zero Differential Attrition

500 randomized (245 Bicarbonate vs 255 Placebo). Only 2 patients withdrew consent (1 in each group). Modified ITT analyzed 244 vs 254 (99.6% complete follow-up).

Infusion Compliance

493/498 (99.0%) received the assigned infusion (243/244 Bicarbonate vs 250/254 Placebo). Median start time was rapid (0.4-0.5h post-randomization).

Minimal Selection Bias

Reasons for non-randomization among 138 eligible patients were logistical (staff unavailability [96], physician refusal [27], patient refusal [15]).

Table 1 (Part 1): Baseline Patient Characteristics

Table 1. Characteristics of the Patients at Baseline (Modified Intention-to-Treat Population). ^{*,‡}		
Characteristic	Sodium Bicarbonate (N = 244)	Placebo (N = 254)
Median age (IQR) — yr	65.0 (51.8 to 74.0)	68.0 (55.2 to 76.0)
Sex — no. (%)		
Female	111 (45.5)	104 (40.9)
Male	133 (54.5)	150 (59.1)
Median time between eligibility determination and randomization (IQR) — min	39.9 (15.0 to 73.3)	37.8 (12.0 to 70.6)
Type of admission — no. (%)		
Medical	119 (48.8)	123 (48.4)
Emergency surgery	64 (26.2)	66 (26.0)
Elective surgery	61 (25.0)	65 (25.6)
Median APACHE II score on ICU admission (IQR)†	21 (16 to 26)	22 (17 to 27)
Median SOFA score closest to randomization (IQR)‡	7 (6 to 9)	7 (6 to 9)
Cause of metabolic acidosis — no. (%)§		
Sepsis or septic shock	122 (50.0)	119 (46.9)
Cardiogenic shock	40 (16.4)	56 (22.0)
Hypovolemia or dehydration	32 (13.1)	20 (7.9)
Cardiac arrest	19 (7.8)	28 (11.0)
Postoperative condition	9 (3.7)	15 (5.9)
Vasoplegia	6 (2.5)	5 (2.0)
Acute kidney injury¶	3 (1.2)	5 (2.0)
Hemorrhage	3 (1.2)	3 (1.2)
Anaphylaxis or anaphylactic shock	1 (0.4)	1 (0.4)
Other	9 (3.7)	2 (0.8)
Blood gas variables at eligibility determination		
Median pH (IQR)	7.26 (7.22 to 7.28)	7.26 (7.23 to 7.28)
pH <7.25 — no. (%)	93 (38.1)	99 (39.0)
Median base excess (IQR) — mmol/liter	−9.2 (−11.5 to −7.0)	−8.8 (−11.0 to −7.0)
Median PaCO ₂ (IQR) — mm Hg	41 (35 to 44)	41 (36 to 44)
Median bicarbonate (IQR) — mmol/liter	17.0 (15.0 to 19.2)	18.0 (15.9 to 19.1)

Table 1 (Part 2): Baseline Severity & Renal Function

.. (Continued.)		
Characteristic	Sodium Bicarbonate (N = 244)	Placebo (N = 254)
Cr creatinine before ICU admission (IQR) — mg/dl	1.05 (0.80 to 1.40)	1.04 (0.81 to 1.41)
Randomization variables		
Cr creatinine (IQR) — mg/dl	1.46 (1.07 to 2.18)	1.42 (1.05 to 2.14)
Cr lactate (IQR) — mmol/liter	2.5 (1.4 to 4.5)	2.5 (1.5 to 4.5)
Cr sodium (IQR) — mmol/liter	137 (133 to 140)	137 (134 to 139)
Received sodium bicarbonate — no. (%)	35 (14.3)	23 (9.1)
Acute kidney injury — no./total no. (%)**	113/243 (46.5)	115/251 (45.8)
Mechanical ventilation — no. (%)	191 (78.3)	207 (81.5)
ARDS — no. (%)††	163 (66.8)	174 (68.5)
Septic shock — no. (%)††	95 (38.9)	104 (40.9)
Cr norepinephrine dose (IQR) — µg/kg/min	0.14 (0.07 to 0.23)	0.15 (0.07 to 0.23)

Modified intention-to-treat population included all the patients who underwent randomization and did not withdraw consent. IQR denotes interquartile range, and Paco₂ partial pressure of arterial carbon dioxide.

Acute Physiology and Chronic Health Evaluation (APACHE) II is a scoring system used in critically ill patients. Scores were calculated on the day of admission to the intensive care unit (ICU). Scores range from 0 to 71, with higher scores indicating a higher severity of illness and a greater likelihood of death in the hospital.

Sequential Organ Failure Assessment (SOFA) score is a scoring system used in critically ill patients. Scores range from 0 to 24, with higher scores indicating a higher severity of illness. The calculation used in this trial excluded the neurologic component of SOFA.

Presumed primary cause of metabolic acidosis was recorded by the bedside clinical team on the basis of the clinical assessment at the time of enrollment.

Acute kidney injury included acute-on-chronic kidney disease.

Cr creatinine level reported here was the most recent serum creatinine level measured within 3 months and more than 7 days before the ICU admission. When data were not available, this creatinine level was imputed according to the following formula: creatinine=0.74–0.2 [if female]+0.08 [if Black]+0.003×age [in years]. To convert the values for creatinine to micromoles per liter, multiply by 88.4.

Cr acute kidney injury before randomization was determined on the basis of the creatinine criteria from the Kidney Disease: Improving Global Outcomes (KDIGO) definition.¹⁹

Cr infection was defined as infection and a total or modified SOFA score of at least 2. Septic shock was defined as sepsis, use of vasopressors, and a lactate level of more than 2 mmol per liter.

Baseline Comparability & Prognostic Balance

Well-Balanced Key Covariates

- **Severity of Illness:**
 - Median APACHE II score: 21 (Bicarbonate) vs 22 (Placebo)
 - Median SOFA score: 7.0 vs 7.0
- **Acid–Base Profile at Entry:**
 - Median pH: 7.26 (IQR 7.22-7.28) vs 7.26 (IQR 7.23-7.28)
 - Severe acidemia (pH < 7.25): 38.1% vs 39.0%
 - Median Base Excess: -9.2 vs -8.8 mmol/L
 - Median PaCO₂: 41 vs 41 mmHg
- **Vasopressor Exposure:**
 - Norepinephrine equivalent: 0.14 vs 0.15 µg/kg/min

Minor Asymmetries & Clinical Nuances

- **Etiology of Acidosis Distribution:**
 - Sepsis/Septic shock: 50.0% vs 46.9%
 - Cardiogenic shock: 16.4% vs 22.0%
 - Cardiac arrest: 7.8% vs 11.0%
- **Pre-randomization Open-label Buffer:**
 - 14.3% in Bicarbonate vs 9.1% in Placebo received open-label NaHCO₃ prior to randomization (adjusted for in sensitivity analyses).
- **Renal Impairment at Baseline:**
 - 46.5% vs 45.8% met KDIGO AKI criteria at entry; baseline creatinine ~1.4 mg/dL.

Table 2: Intervention Delivery & Acid-Base Evolution

Table 2. Characteristics of the Intervention in the First 24 Hours.			
Characteristic	Sodium Bicarbonate (N = 244)	Placebo (N = 254)	Adjusted Absolute Difference (95% CI)*
Median time between randomization and start of infusion (IQR) — hr	0.5 (0.4 to 0.7)	0.4 (0.3 to 0.5)	0.1 (0.1 to 0.2)
Median duration of infusion (IQR) — hr	5.0 (4.7 to 5.0)	5.0 (5.0 to 5.0)	0.0 (0.0 to 0.0)
Median volume infused (IQR) — ml	500 (329 to 500)	500 (500 to 500)	0.0 (−32.0 to 32.0)
Median total sodium bicarbonate dose (IQR) — mmol	300.0 (197.4 to 300.0)	0.0 (0.0 to 0.0)	300.0 (292.4 to 307.6)
Adjustment of infusion — no./total no. (%)†	121/243 (49.8)	58/250 (23.2)	27.0 (19.5 to 34.5)
Use of open-label sodium bicarbonate — no./total no. (%)	17/243 (7.0)	38/250 (15.2)	−8.2 (−13.6 to −2.8)
Blood gas analysis during the first 24 hr‡			
Median pH (IQR)			
Lowest	7.31 (7.26 to 7.34)	7.26 (7.22 to 7.30)	0.05 (0.04 to 0.06)
Highest	7.43 (7.39 to 7.46)	7.37 (7.33 to 7.41)	0.05 (0.04 to 0.06)
Median bicarbonate (IQR) — mmol/liter			
Lowest	19.0 (17.0 to 22.0)	17.0 (15.0 to 19.0)	2.6 (1.7 to 3.4)
Highest	26.0 (23.8 to 27.9)	21.0 (19.0 to 23.0)	4.5 (3.8 to 5.2)
Median base excess (IQR) — mmol/liter			
Lowest	−6.0 (−9.0 to −3.5)	−9.0 (−12.0 to −7.0)	3.1 (2.4 to 3.7)
Highest	1.5 (−1.0 to 3.0)	−4.2 (−6.2 to −1.8)	5.1 (4.4 to 5.8)
Median Paco ₂ (IQR) — mm Hg			
Lowest	35.0 (32.0 to 38.9)	33.9 (30.0 to 37.0)	1.0 (−0.2 to 2.2)
Highest	44.0 (40.0 to 48.0)	41.8 (38.0 to 46.0)	2.0 (0.6 to 3.4)
Median lactate (IQR) — mmol/liter			
Lowest	1.3 (1.0 to 2.2)	1.3 (0.9 to 1.9)	0.1 (0.0 to 0.2)
Highest	3.1 (2.1 to 5.4)	2.9 (1.8 to 5.1)	0.2 (−0.1 to 0.5)
Median sodium (IQR) — mmol/liter			
Lowest	136.5 (134.0 to 140.0)	133.0 (131.0 to 136.0)	4.0 (2.6 to 5.4)
Highest	141.0 (138.0 to 144.0)	137.0 (134.0 to 140.0)	4.0 (2.6 to 5.4)
Median norepinephrine infusion rate (IQR) — µg/min			
Lowest	0.06 (0.03 to 0.10)	0.05 (0.02 to 0.12)	0.00 (−0.01 to 0.01)
Highest	0.18 (0.10 to 0.31)	0.19 (0.11 to 0.30)	−0.01 (−0.04 to 0.02)
* Differences were adjusted for stratification factors (pH and creatinine levels), with site included as a random effect. Absolute differences between percents are shown in percentage points; other absolute differences are shown in the units specified in the table.			
† Adjustment to the infusion was defined as any protocol-driven change in the infusion rate from the initial 100 ml per hour during the 5-hour trial infusion, on the basis of prespecified assessments at 1 hour and 3 hours after the start of the infusion.			
‡ This variable was derived from measurements obtained at six time points during the first 24 hours after randomization and initiation of the infusion (at 1, 3, 5, 12, 18, and 24 hours). Sodium bicarbonate or placebo was infused until hour 5.			

Table 3: Primary & Secondary Clinical Trial Outcomes

Table 3. Trial Outcomes.*				
Outcome	Sodium Bicarbonate (N = 244)	Placebo (N = 254)	Adjusted Absolute Difference (95% CI)	Adjusted Hazard Ratio (95% CI)
Primary outcome				
Major adverse kidney event within 30 days — no. (%)	98 (40.2)	100 (39.4)	1.2 (−7.1 to 9.4)†	—
Secondary outcomes				
Components of primary outcome‡:				
In-hospital death by day 30 — no. (%)	62 (25.4)	61 (24.0)	1.8 (−5.6 to 9.2)	1.09 (0.76 to 1.56)
Receipt of renal-replacement therapy within 30 days — no. (%)	41 (16.8)	53 (20.9)	−3.9 (−10.6 to 2.7)	—
Persistent renal dysfunction within 30 days — no./total no. (%)§	34/243 (14.0)	46/251 (18.3)	−4.3 (−10.7 to 2.1)	—
Renal-replacement therapy dependence at day 30 — no. (%)¶	14 (5.7)	23 (9.1)	−3.3 (−7.8 to 1.3)	—
In-ICU death by day 30 — no. (%)	49 (20.1)	56 (22.0)	−1.7 (−8.6 to 5.3)	—
In-hospital death by day 90 — no. (%)	64 (26.2)	65 (25.6)	1.2 (−6.3 to 8.6)	1.06 (0.75 to 1.51)
Physiological outcomes				
Recurrence of metabolic acidosis — no./total no. (%)	78/244 (32.0)	141/253 (55.7)	−23.3 (−31.4 to −15.1)	—
Acute kidney injury — no./total no. (%)**	156/242 (64.5)	162/250 (64.8)	−0.0 (−7.7 to 7.6)	—
Stage 1	64/156 (41.0)	58/162 (35.8)		
Stage 2	35/156 (22.4)	44/162 (27.2)		
Stage 3	57/156 (36.5)	60/162 (37.0)		
Process-of-care measures††				
Median no. of vasopressor-free days at day 30 (IQR)	27.2 (0.0 to 28.9)	27.6 (6.1 to 28.9)	−0.5 (−1.3 to 0.4)	
Median no. of days free from renal-replacement therapy at day 30 (IQR)	30.0 (0.0 to 30.0)	30.0 (10.4 to 30.0)	−0.0 (−0.6 to 0.6)	
Median no. of ICU-free days at day 30 (IQR)	23.7 (0.0 to 26.9)	24.0 (0.0 to 27.0)	−0.5 (−2.3 to 1.3)	
Median no. of hospital-free days at day 90 (IQR)	69.3 (0.0 to 79.8)	69.2 (0.0 to 80.6)	1.1 (−6.4 to 8.6)	
* Analyses were adjusted for stratification factors (pH and creatinine levels), with site included as a random effect. Absolute differences between percents are shown in percentage points; other absolute differences are shown in the units specified in the table. The widths of the confidence intervals for secondary outcomes have not been adjusted for multiple comparisons, so the intervals should not be used to infer definitive differences between groups.				
† P=0.78.				
‡ Components of the primary composite outcome were not mutually exclusive; some participants contributed to more than one component.				
§ Persistent renal dysfunction within 30 days was defined as a final inpatient creatinine value, occurring closest to day 30, of at least 200% of the baseline value.				
¶ Renal-replacement therapy dependence at day 30 was defined as the receipt of any form of renal-replacement therapy within a window of ±10 days of the 30-day time point after randomization.				
Recurrence of metabolic acidosis was defined as meeting the criteria for metabolic acidosis (pH, <7.30; base excess of no more than −4 mmol per liter; and Paco ₂ , ≤45 mm Hg) during the first 7 days after randomization, including during the trial infusion period. Acidosis must have first resolved before a subsequent episode could be classified as a recurrence.				
** Determination of acute kidney injury was based on the creatinine criteria from the KDIGO definition ¹⁹ as assessed in the first 7 days after randomization. When not available, the creatinine level before ICU admission was imputed according to the following formula: creatinine=0.74−0.2 [if female]+0.08 [if Black]+0.003×age [in years].				
†† The process-of-care measures as assessed in this trial were defined as 30 or 90 days, as specified, of receipt of a continuous infusion of vasopressor or renal-replacement therapy or as days free from the ICU or hospital. Patients who had died by day 30 or 90 received a value of 0.				

Primary Endpoint Analysis: MAKE30 & Components

Comprehensive Outcome Breakdown at Day 30

Primary Composite: MAKE30	98 / 244 (40.2%)	100 / 254 (39.4%)	+1.2% (-7.1 to 9.4)
• In-hospital Death by Day 30	62 / 244 (25.4%)	61 / 254 (24.0%)	+1.8% (-5.6 to 9.2)
• Receipt of RRT within 30 days	41 / 244 (16.8%)	53 / 254 (20.9%)	-3.9% (-10.6 to 2.7)
• Persistent Renal Dysfunction	34 / 243 (14.0%)	46 / 251 (18.3%)	-4.3% (-10.7 to 2.1)
• RRT Dependence at Day 30	14 / 244 (5.7%)	23 / 254 (9.1%)	-3.3% (-7.8 to 1.3)
• Recurrence of Metabolic Acidosis	78 / 244 (32.0%)	141 / 253 (55.7%)	-23.3% (-31.4 to -15.1)

Figure 2A & 2B: In-Hospital Survival & RRT Free Survival

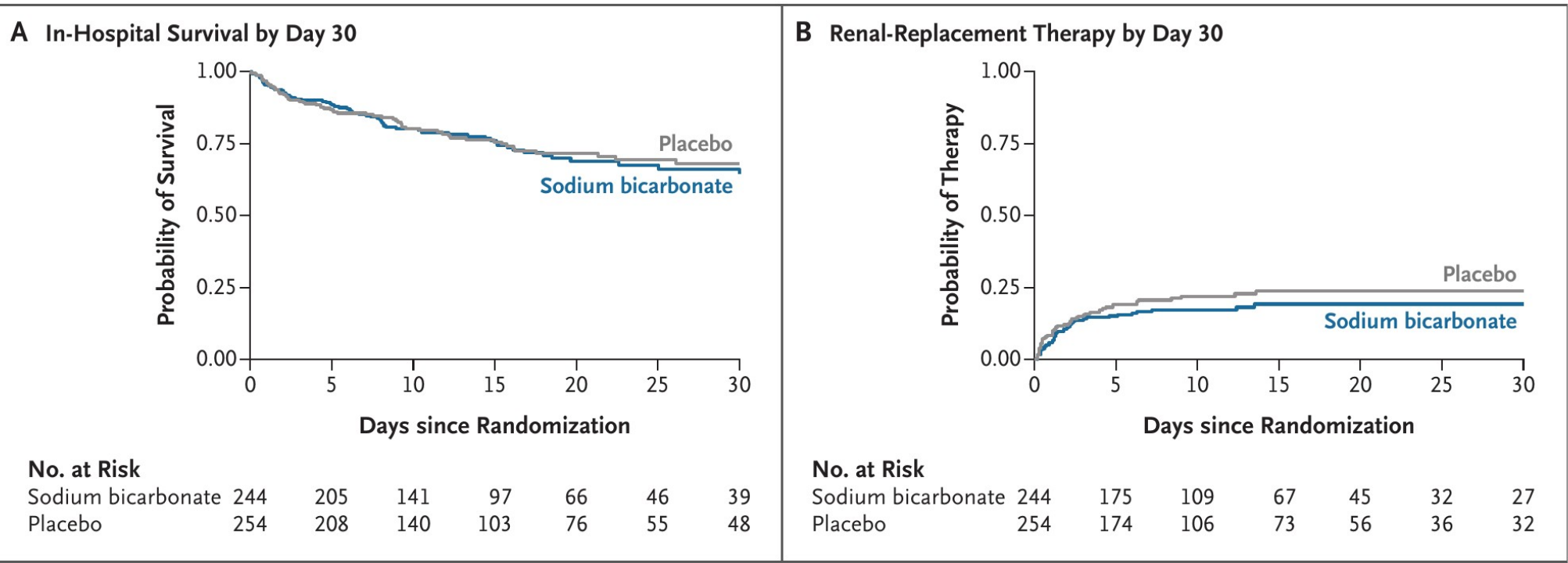
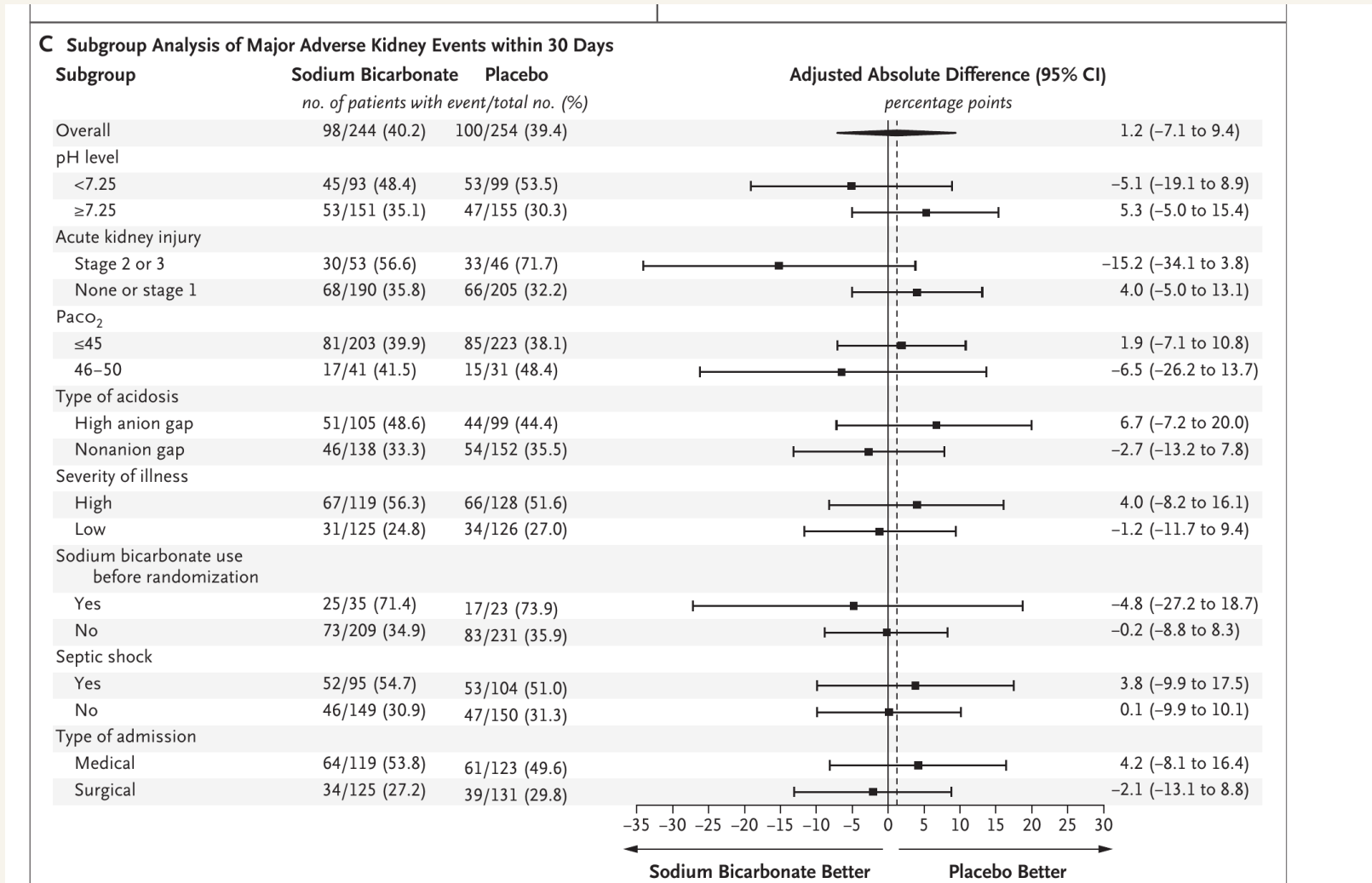


Figure 2C: Prespecified Subgroup Analyses Forest Plot



Subgroup Appraisal: Was There Any Latent Benefit?

Evaluation of Key Stratified Subgroups

1. Severe Acidosis (pH < 7.25):

- Event rate: 48.4% (45/93) in Bicarb vs 53.5% (53/99) in Placebo.
- Adj Diff: -5.1% pts (95% CI: -19.1 to 8.9) — P-for-interaction = NS.

2. Advanced AKI (Stage 2 or 3):

- Event rate: 56.6% (30/53) in Bicarb vs 71.7% (33/46) in Placebo.
- Adj Diff: -15.2% pts (95% CI: -34.1 to 3.8) — wide CI crossing zero.

3. High vs Low Anion Gap Acidosis:

- High AG: +6.7% (-7.2 to 20.0); Non-AG: -2.7% (-13.2 to 7.8).

Critical Appraisal of Subgroup Findings

• No Statistically Significant Interaction:

- Formal interaction tests across all 8 prespecified subgroups showed NO significant heterogeneity of treatment effect.

• Power Limitations in Subgroups:

- AKI Stage 2/3 subgroup had only 99 patients; the apparent 15% absolute point difference carries massive imprecision.

• Takeaway:

- Subgroup trends are purely hypothesis-generating and do NOT justify routine buffer administration even in AKI stage 2/3 shock.

Adverse Effects & Physiological Safety Profile

Observed Safety Outcomes & Electrolyte Disturbances

ADVERSE EFFECTS

1.6% vs 0.0%

4 patients in Bicarbonate group experienced prespecified adverse effects vs 0 in Placebo (P=0.06). Driven entirely by severe hypokalemia requiring urgent potassium replacement.

ELECTROLYTE SHIFTS

Hypokalemia & Hypernatremia

Bicarbonate causes intracellular potassium shifting (H+/K+ exchange) and delivers 300 mmol Na+ load. Median sodium reached 141 mmol/L vs 137 mmol/L.

FLUID BALANCE

500 mL Load

Total median infusion volume was identical (500 mL in both arms). Total cumulative fluid balance through Day 7 was similar across groups.

OPEN-LABEL ESCALATION

7.0% vs 15.2%

Open-label 'rescue' bicarbonate was given in 7.0% of Bicarbonate arm vs 15.2% of Placebo arm (Adj Diff -8.2%, 95% CI -13.6 to -2.8), reflecting clinician rescue.

Critical Appraisal: Internal Validity & Strengths

Key Methodological Strengths

1. Rigorous Double-Blinding:

- Unlike prior open-label trials (BICAR-ICU), SODa-BIC successfully masked clinicians, avoiding bias in RRT initiation decisions.

2. Excellent Allocation Concealment:

- Centralized web-based randomization with independent solution preparation.

3. Near-Zero Loss to Follow-up:

- 99.6% complete primary outcome ascertainment (only 2 consent withdrawals).

4. Pragmatic Protocolized Titration:

- Rapid start (median 0.4h), clear physiological stop-criteria at 1h, 3h, and 5h.

Methodological Vulnerabilities & Caveats

1. Fixed 5-Hour Infusion Horizon:

- Infusion capped at 5 hours; whether prolonged or repeated buffer infusions in refractory shock alter outcomes remains unaddressed.

2. Potential Unblinding via ABG:

- Rapid pH and bicarbonate rise on ABG may have allowed astute clinicians to deduce group assignments.

3. Open-Label Contamination (15.2% in Placebo):

- Post-intervention rescue buffer use in placebo may have diluted between-group effect sizes.

4. D5W Placebo Tonicity:

- 5% dextrose is hypotonic compared to 600 mmol/L NaHCO₃ (unavoidable for visual masking).

Cochrane Risk of Bias 2 (RoB 2) Assessment

RoB 2 Domain-by-Domain Evaluative Summary

Domain 1: Randomisation Process	LOW RISK	Centralized web randomization, permuted blocks, stratified by center, pH, and creatinine. Baseline prognostic balance well maintained.
Domain 2: Deviations from Intended Interventions	LOW RISK	Double-blind design with matched D5W solutions. 99% adherence to 5h protocol. Pre-specified sensitivity analyses adjusted for open-label buffer.
Domain 3: Missing Outcome Data	LOW RISK	Primary endpoint ascertained in 498/500 (99.6%). Only 2 consent withdrawals with zero differential loss to follow-up.
Domain 4: Measurement of Outcome	LOW RISK	MAKE30 components (mortality, RRT initiation, creatinine $\geq 200\%$) objective and adjudicated blindly.
Domain 5: Selection of Reported Result	LOW RISK	Published protocol and pre-specified Statistical Analysis Plan (SAP). All primary/secondary outcomes reported transparently.
OVERALL RISK OF BIAS	LOW RISK	High methodological quality RCT providing definitive, unbiased evidence on sodium bicarbonate in ICU shock.

GRADE Certainty of Evidence Summary

GRADE Evidence Profile for Sodium Bicarbonate in Shock with Acidosis

MAKE30 (Day 30)	RoB: Low Incon: Not Serious Indir: Not Serious	HIGH	Adequately powered trial; CI excludes meaningful benefit.
All-Cause Mortality (Day 30)	RoB: Low Incon: Not Serious Indir: Not Serious	HIGH	HR 1.09 (0.76-1.56); zero mortality reduction.
Need for RRT (Day 30)	RoB: Low Incon: Not Serious Indir: Not Serious	HIGH	16.8% vs 20.9% (Diff -3.9%, 95% CI -10.6 to 2.7).
Vasopressor-Free Days	RoB: Low Incon: Not Serious Indir: Not Serious	HIGH	Median 27.2 vs 27.6 days; no hemodynamic benefit.
Adverse Events (Hypokalemia)	RoB: Low Incon: Not Serious Indir: Not Serious	MODERATE	Low event rate (4 vs 0); physiologically expected.

Chronology of Landmark Trials: Sodium Bicarbonate in ICU

Evolution from Open-Label Signals to Definitive Double-Blind Rebuttal

BICAR-ICU (Lancet 2018)

Open-Label Phase 3

N = 389, pH $\leq$ 7.20 (Severe acidemia)

Primary composite (28d death/organ failure): 66% vs 71% (P=0.35). Subgroup with AKI: Mortality 46% vs 63% (P=0.02), RRT 51% vs 73% (P=0.002).

BICARICU-2 (JAMA 2025)

Open-Label RCT in AKI

N = 300, pH $\leq$ 7.20 + KDIGO AKI

90-day mortality: 45% vs 48% (P=0.62). Reduced RRT use (42% vs 58%, P=0.008). Open-label bias suspected in RRT timing.

SODa-BIC Pilot (CCM 2023)

Double-Blind Pilot

N = 40, Vasopressor Shock, pH < 7.30

Feasibility confirmed: rapid pH normalization within 5h, less acidosis recurrence, no safety signals. Set up the definitive trial.

SODa-BIC (NEJM 2026)

Double-Blind, Placebo-Controlled

N = 500, Vasopressor Shock, pH < 7.30

MAKE30: 40.2% vs 39.4% (Adj Diff +1.2%, P=0.78). 30d Death: 25.4% vs 24.0%. RRT: 16.8% vs 20.9%. Vasopressor duration unchanged.

Guideline Consensus & Critical Care Standards

Surviving Sepsis Campaign (SSC 2021)

- **Recommendation 46 (Weak Recommendation, Low Quality Evidence):**

"For adults with septic shock and hypoperfusion-induced lactic acidemia, we suggest AGAINST the use of sodium bicarbonate therapy to improve hemodynamics or reduce vasopressor requirements if $\text{pH} \geq 7.20$."

- **Recommendation 47 (Weak, Low Quality Evidence):**

"For adults with septic shock, severe metabolic acidemia ($\text{pH} \leq 7.20$) and AKI (AKIN score 2-3), we suggest using sodium bicarbonate therapy."

- **SODa-BIC Impact:**

Directly challenges routine sodium bicarbonate use in shock with pH 7.20-7.30 and questions AKI subgroup benefit.

KDIGO & French Intensive Care Society

- **KDIGO AKI Guidelines (2012 / Updates):**

- No recommendation for routine bicarbonate to prevent AKI or postpone RRT in general ICU populations.
- Emphasizes treatment of underlying etiology, fluid optimization, and vasopressor support.

- **SFAR / SRLF Guidelines (2019):**

- Suggested NaHCO_3 only in severe metabolic acidemia ($\text{pH} \leq 7.20$, bicarbonate ≤ 20 mmol/L) with AKI stage 2-3 based on BICAR-ICU.

- **Summary Consensus:**

- SODa-BIC provides definitive evidence that pH correction alone does NOT improve renal recovery or survival.

Local Applicability in North Indian / Ghaziabad ICU Practice

Resource-Constrained ICU Realities

1. High Baseline Sepsis & Tropical Shock Burden:

- Patients frequently present late with septic shock (pneumonia, urosepsis, tropical infections like scrub typhus, dengue).
- Severe metabolic acidosis ($\text{pH} < 7.25$, high lactate) is ubiquitous.

2. Widespread Empirical Bicarbonate Overuse:

- 8.4% NaHCO_3 ampoules are frequently 'pushed' empirically to 'buy time' or improve blood pressure.

3. Dialysis Access & Financial Constraints:

- CRRT is limited by cost; intermittent hemodialysis/SLED requires vascular access and hemodynamic stability.

Clinical Implications for Santosh Hospital

• Stop Routine 'Cosmetic' Buffer Therapy:

- Normalizing numbers on an ABG machine does not save nephrons or lives.

• Prioritize Etiological & Hemodynamic Resuscitation:

- Focus efforts on source control, timely broad-spectrum antibiotics, balanced crystalloids, and titrated vasopressors (Norepinephrine + Vasopressin).

• Monitor Electrolytes Closely:

- Bicarbonate infusions risk severe hypokalemia and hypocalcemia in malnourished, septic Indian patients.

Actionable Takeaways across Clinical Environments

EMERGENCY & WARD

- Do not give empirical bicarbonate boluses for lactic acidosis or sepsis with pH 7.20-7.30.
- Rapidly identify septic focus, start culture-guided antibiotics, and restore organ perfusion.
- Reserve bicarbonate for true bicarbonate loss (severe diarrhea, type 2 RTA) or toxicological indications (TCA toxicity, severe hyperkalemia).

ICU & SHOCK MANAGEMENT

- SODa-BIC proves that 5-hour sodium bicarbonate infusion does NOT reduce vasopressor needs or 30-day MAKE30.
- Treat the shock, not the pH. Inotrope responsiveness is restored by resolving tissue hypoperfusion, not buffer infusions.
- Watch for iatrogenic hypokalemia and hyponatremia if bicarbonate is infused.

NEPHROLOGY & RRT PLANNING

- Deferral of RRT seen in open-label trials was largely driven by pH-threshold masking.
- Base RRT decisions on refractory hyperkalemia, volume overload, severe uremia, or worsening refractory acidosis (pH < 7.15) rather than relying on bicarbonate to avert dialysis.

Resident Journal Club Discussion Questions

Q1: Biological vs Clinical Discordance

Why did sodium bicarbonate successfully normalize arterial pH and base excess within 5 hours, yet fail completely to reduce vasopressor requirements, AKI progression, or mortality?

Q2: The Open-Label Trap

How did the double-blind design of SODa-BIC expose the confirmation bias present in BICAR-ICU regarding RRT initiation thresholds?

Q3: The 5-Hour Duration Dilemma

Was capping the infusion at 5 hours a methodological flaw, or did it accurately test whether early buffer correction alters the downstream shock trajectory?

Q4: Severe Acidosis (pH < 7.15-7.20)

Does SODa-BIC completely close the door on bicarbonate in ultra-severe acidemia (pH < 7.15), or is there still clinical equipoise in profound metabolic acidosis with AKI?

Critical Appraisal: What Would Change My Mind?

Future Evidence Requirements to Re-Evaluate Buffer Therapy

Profound Acidemia RCT (pH < 7.15)

A double-blind, placebo-controlled trial powered specifically for ultra-severe acidemia (pH < 7.15) with Stage 3 AKI, where cellular enzymatic dysfunction is maximal.

Continuous / Extended Buffer Delivery

A trial evaluating continuous titration of buffer therapy beyond 5 hours until complete resolution of shock and hyperlactatemia.

Non-CO2 Generating Alternative Buffers

Comparative evaluation of buffers that do not generate carbon dioxide (such as THAM - Tris-hydroxymethyl aminomethane or Carbicarb) to prevent paradoxical intracellular acidosis.

Standardized Biological AKI Biomarkers

Studies utilizing novel renal tubular damage biomarkers (TIMP-2 * IGFBP7, NGAL, urinary clusterin) rather than serum creatinine to detect subclinical nephron salvage.

Key References & Relevant Literature

References & Methodological Standards

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Sodium Bicarbonate in ICU Shock: The Final Verdict

- **Key Takeaway:** In this definitive double-blind RCT (N=500), sodium bicarbonate normalized pH faster but provided ZERO clinical benefit for 30-day MAKE30, RRT avoidance, vasopressor duration, or survival.
- **Practice Change:** Do not administer routine sodium bicarbonate for metabolic acidosis with pH < 7.30 in shock.
- **Clinical Principle:** Treat the underlying cause of shock and hypoperfusion. Correcting the number on an ABG is cosmetic, not curative.

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